Homework 6

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27-03-2025

Question 1:

(Grad version) Reproduce this plot as closely as you can. Hint: Read the plotmath documentation, and consider the construction expression(paste(. . .)).

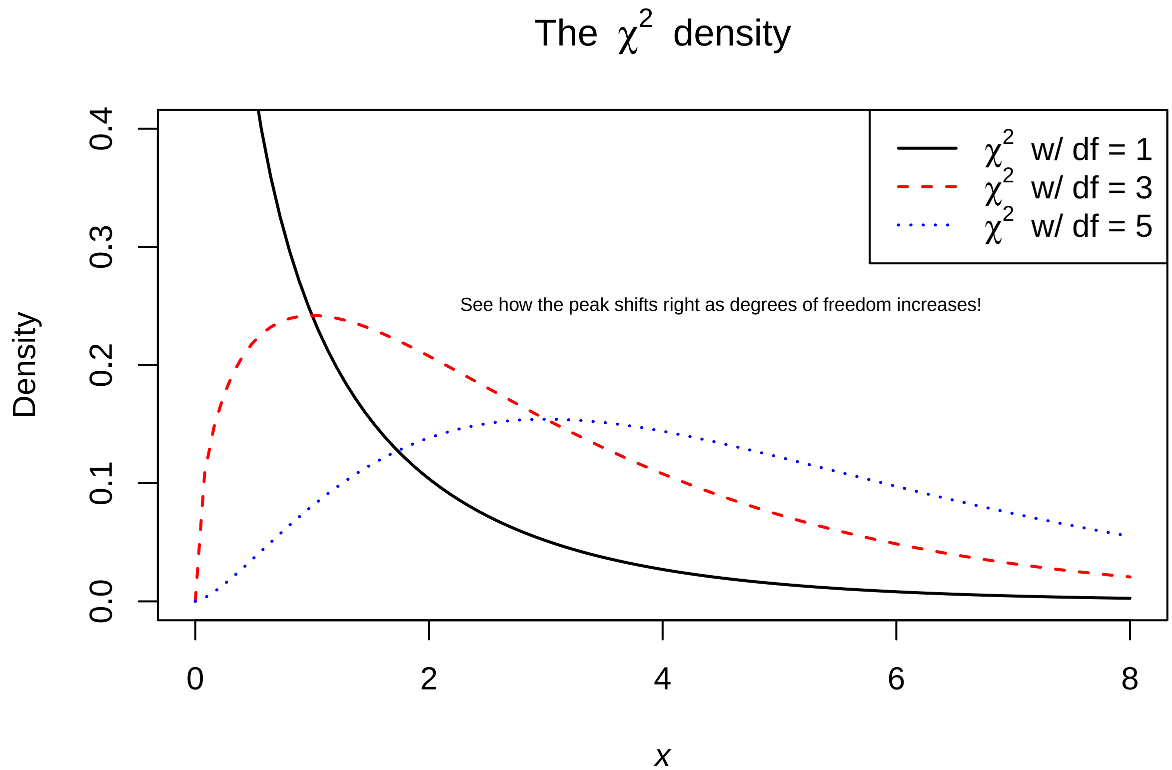
```
# Set up the x-axis values
x <- seq(0, 8, length.out = 100)

# Plot the first chi-square density curve
plot(x, dchisq(x, df = 1), type = "l", lwd = 1.5, col = "black",
     xlab = expression(italic(x)), ylab = "Density",
     main = expression("The " ~ chi^2 ~ " density"),
     ylim = c(0, 0.4))

# Add additional density curves
lines(x, dchisq(x, df = 3), col = "red", lwd = 1.5, lty = 2)
lines(x, dchisq(x, df = 5), col = "blue", lwd = 1.5, lty = 3)

# Add a legend
legend("topright", legend = c(expression(chi^2 ~ " w/ df = 1"),
                              expression(chi^2 ~ " w/ df = 3"),
                              expression(chi^2 ~ " w/ df = 5")),
       col = c("black", "red", "blue"), lty = c(1, 2, 3), lwd = 1.5)

# Add annotation
text(4.5, 0.25, "See how the peak shifts right as degrees of freedom increases!", cex = 0.6)
```



Question 2:

eproduce this plot as closely as you can. I'll get you started with the first two lines that produce the data. Create the appropriate formula objects to pass into lm(), and extract the $fitted.values object to plot the prediction curve. Hint: To avoid repetition in your code, can you use lapply() to iterate over degrees and colors?

```r
# Generate data
x <- seq(1, 10, 0.5)
y <- sin(x)

# Scatter plot of original data
plot(x, y, col = "orange", pch = 1, ylim = c(-1, 1),
     main = "Polynomial approximation of sin(x)",
     xlab = "x", ylab = "y")

# Degrees and corresponding colors
degrees <- 1:5
colors <- c("red", "blue", "green", "orange", "gray")

# Fit models and add lines
fits <- lapply(degrees, function(d) {
  model <- lm(y ~ poly(x, d, raw = TRUE))
  lines(x, model$fitted.values, col = colors[d], lwd = 2)
```

```r
  return(model)
})

# Add legend
legend("top",
       legend = paste("Degree", degrees),
       col = colors, lwd = 2, bty = "n", cex = 0.8)
```